0-13

Algorithm
↓
free gate 2025
↓
Pyqs
↓
practice

Lec_14

Complexity Classes

Type of functions

(+ve functions)

$f(n) = n^3$, $f(n) = n$, $f(n) = \log_2 n$

$f(n) = \log_2 \log_2 n$ $f(n) = 2^n$

$f(n) = \log_4 n$

Ex:- $\frac{1}{n}, \frac{1}{n^2}, \frac{1}{n^3}, \frac{1}{\log n}, \frac{1}{\log\log n}$

$f(n) = n^3$

1.) Decreasing functions

Increasing $\Rightarrow$ $n\uparrow$ $f(n)\uparrow$

$n = 2$, $f(2) = 2^3 = 8$

$n = 3$, $f(3) = 3^3 = 27$

$n = 4$, $f(4) = 4^3 = 64$

decreasing $n\uparrow$ $f(n)\downarrow$

Ex:- 1) $f(n) = 1/n$

$n = 1$, $f(1) = 1$

$n = 2$, $f(2) = 1/2$

$n = 3$, $f(3) = 1/3$

2) $f(n) = 1/n^4$

$n = 1$, $f(1) = \frac{1}{1^4} = 1$

$n = 2$, $f(2) = \frac{1}{2^4} = \frac{1}{16}$

$n = 3$, $f(3) = \frac{1}{3^4}$

2.) Constant functions

2, 3, 4, 5

$f(n) = 2n^0, 3n^0, 4n^0, 5n^0$

3.)Logarthmic Functions :

$\log_2 n$, $\log_4 n$, $\log_2 \log_3 n$, $(\log_2 n)^2$

4.) Linear functions :

⇒ power 1

$f(n) = n$, $f(n) = 2n$, $f(n) = n+2$

→ two

5.)Quadratic functions

Ex:

$f(n) = n^2$, n^2+n+2

$f(n) = 3n^2$

6.)Cubic functions

n^3 ③

$f(n) = n^3$, $n^3 + 2$

7.)Polynomial functions

const $\Rightarrow$ 0
Linear $\Rightarrow$ 1
quad. $\Rightarrow$ 2
cubic $\Rightarrow$ 3

$$f(n) = a^n + b^{n-1} + c^{n-2} \cancel{+} 4$$

$\rightarrow e^x$

8.) Exponential functions:

$f(n) = 2^2, 2^{2n}, 3^n, a^n$

$e^n =$

* very Imp concept for the gate

Comparision of Functions

1.) n^2 vs n^3

$n^3 > n^2$ ✓

n^2	n^3
$\cancel{n^2}$	$\cancel{n^2} \cdot n^1$
1	n

2.) $\log_2 n$ vs n

$\log n$	n

$n = 1$, $\log_2(1) = 0$

$n = 2$, $\log_2 2 = 1$

$n = 4$, $\log_2 2^2 = 2$

$n = 2^{20}$, $\log_2 2^{20} = 20$

$\log_2 n$	n
0	1
1	2
2	4
20	2^{20}

$n > \log n$

3.) logn vs n^1/2

Method ⇒ when you are in trap
⇓
take log

	$\log n$	$\sqrt{n}$
$n=1$	0	1
$n=2$	1	$\frac{1}{\sqrt{2}} \Rightarrow 1.21$
$n=3$		

$\log n \mid \sqrt{n}$ → taking log

$\log\log n \mid \log\sqrt{n}$

$\log\log n \mid \frac{1}{2}\log n$

$n = 2^{20}$

$\log n = \log_2 2^{20}$
$= 20 \times \log_2 2$
$= 20$

$\log\log n$
$\log(\log_2 2^{20})$
$\log_2(20) \Rightarrow 4.5$

$\sqrt{n} > \log n$

4.) n vs (logn)3

$n \mid (\log n)^3$

$\log n \mid \log(\log n)^3$

$\log n \mid 3 \log \log n$

$n \mid \log n$

$\Downarrow$

$n > \log n$

$\log n > \log \log n$

$n > (\log n)^3$

5.

2^n | n^2

$\log_2 2^n$ | $\log_2 n^2$

$\Rightarrow n \log_2 2$ | $2 \log_2 n$

$\Rightarrow n \times 1$ | $2 \log n$

n vs $\log n$

$n = 1$

n vs $\log n$

$n > \log n$

$n = 100$

$2^n > n^2$

$2^{100} > (10)^2$

$1024 > 100$

6.

n | $(\log n)^{100}$ | $\log n > \log\log n$

$\log_2 n$ | $100 \log(\log n)^{100}$ | $n > (\log n)^{100}$

$\log_2 n$ | $100 \log\log n$

Thanks Do Subscribe